Project Report: Quiz Master Application

Name: RAHUL

Student id: 23f3002149@ds.study.iitm.ac.in

1. Introduction The Quiz Master Application is a multi-user platform designed to facilitate quiz-based learning. It provides functionalities for both students and administrators, allowing seamless quiz management, score tracking, and data visualization. The project aims to offer an interactive and engaging way for users to test their knowledge on various subjects.

2. Project Overview

- **Title:** Quiz Master Application
- **Description:** A web-based application enabling students to take quizzes and allowing admins to manage quiz content.
- **User & Admin Roles:**
    - **Admin:** Manages subjects, chapters, quizzes, and user data.
    - **Students:** Participate in quizzes and track their performance.

3. Technology Stack

- **Backend:** Flask (Python-based web framework for handling server-side logic)
- **Frontend:** Jinja2 templates (for dynamic HTML generation and styling)
- **Database:** Flask-SQLAlchemy (for data storage and management)
- **Visualization:** Matplotlib (for generating graphical reports of quiz performance)

4. Functionalities

- **User Authentication:** Secure login and role-based access control.
- **Quiz Management:** Admins can create, update, and delete quizzes for different subjects and chapters.
- **Search Functionality:** Users can search quizzes based on subject, chapter,date.

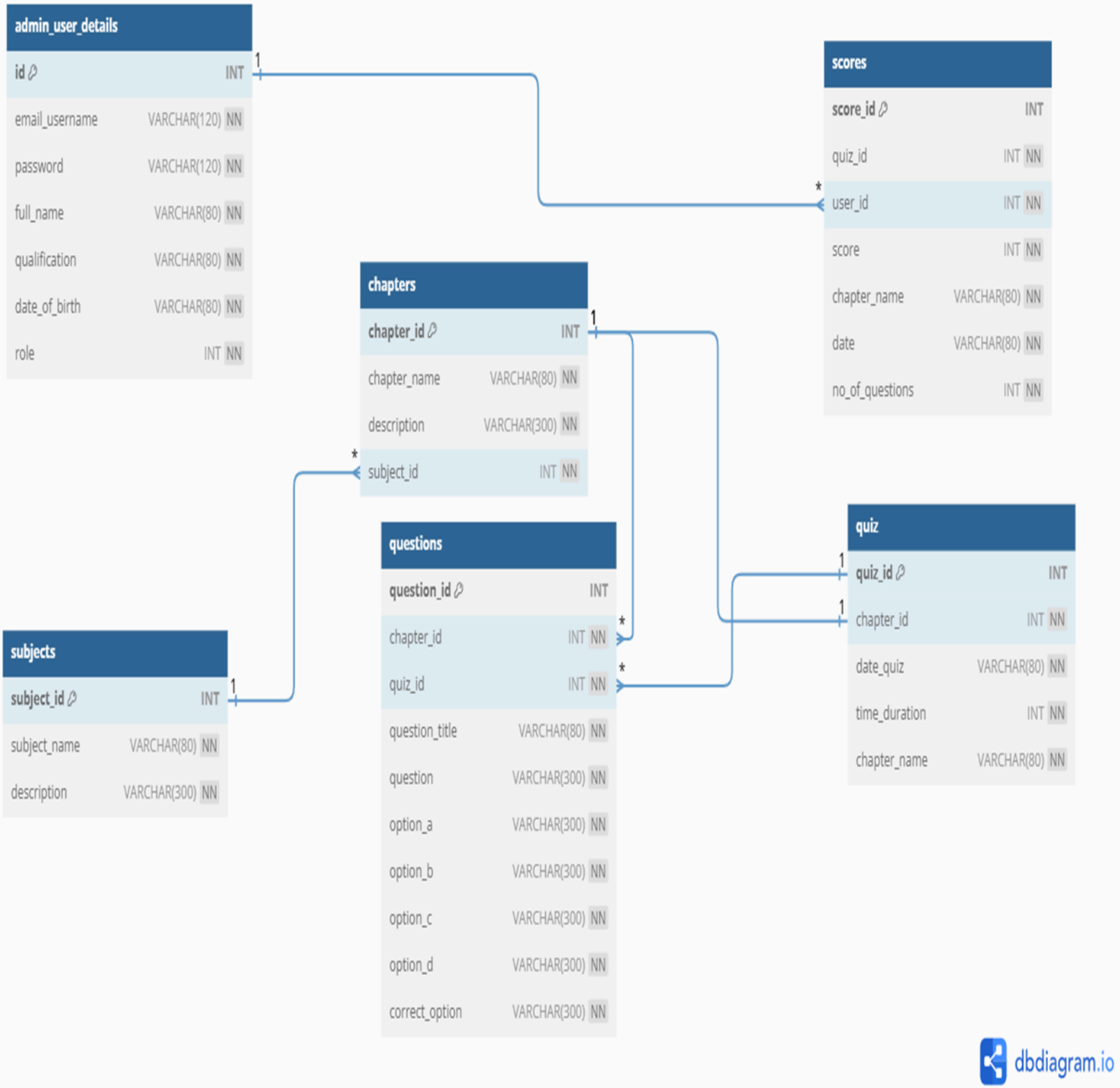
- **Quiz Attempting:** Students can attempt quizzes and get their scores immediately .
- **Performance Tracking:** Users can view their past quiz scores and compare progress.
- **Data Visualization:** Matplotlib is used to generate performance graphs.

5. Development Process & Challenges

- **Initial Phase:** Created HTML pages based on wireframe designs and gradually enhanced them with CSS.
- **Backend Development:** Implemented Flask routes and connected them to respective HTML templates.
- **Challenges Faced:**
    - Difficulty in CSS styling, resolved through online resources like W3Schools.
    - Database model relationships were initially complex, but resolved through SQLAlchemy documentation.
    - Implementing role-based access control required additional learning and testing.

6. Database Design (E-R Diagram) The database consists of several key entities:

- **Admin Users:** Stores admin credentials and personal details.
- **Subjects:** Contains different subjects available for quizzes.
- **Chapters:** Divisions under each subject.
- **Quizzes:** List of quizzes linked to specific chapters.
- **Questions:** Stores individual quiz questions.
- **Scores:** Tracks student quiz submissions and scores.


admin_user_details
id INT
email_username VARCHAR(120) NN
password VARCHAR(120) NN
full_name VARCHAR(80) NN
qualification VARCHAR(80) NN
date_of_birth VARCHAR(80) NN
role INT NN
scores
score_id INT
quiz_id INT NN
user_id INT NN
score INT NN
chapter_name VARCHAR(80) NN
date VARCHAR(80) NN
no_of_questions INT NN
chapters
chapter_id INT
chapter_name VARCHAR(80) NN
description VARCHAR(300) NN
subject_id INT NN
questions
question_id INT
chapter_id INT NN
quiz_id INT NN
question_title VARCHAR(80) NN
question VARCHAR(300) NN
option_a VARCHAR(300) NN
option_b VARCHAR(300) NN
option_c VARCHAR(300) NN
option_d VARCHAR(300) NN
correct_option VARCHAR(300) NN
quiz
quiz_id INT
chapter_id INT NN
date_quiz VARCHAR(80) NN
time_duration INT NN
chapter_name VARCHAR(80) NN
subjects
subject_id INT
subject_name VARCHAR(80) NN
description VARCHAR(300) NN
dbdiagram.io

7. Conclusion The Quiz Master Application successfully creates an interactive quiz-based learning platform. It provides essential functionalities such as authentication, quiz management, and score tracking, ensuring an engaging experience for users. Future enhancements may include AI-based quiz recommendations and a leaderboard system to boost competitiveness.

8. Presentation link:

- **Quiz_Master_App_Presentation.mp4**

8. Future Scope

- **Leaderboard for competitive ranking among students.**
- **Real-time quiz events and challenges.**
- **Mobile application version for better accessibility.**